

Unit 1: Introduction to Biochemistry

Subject:
Biochemistry

Unit 2: Enzymes

System:
Enzymes

The recent advent of multidrug-resistant *Mycobacterium tuberculosis* (Mtb) has led to the search for novel agents to combat tuberculosis. Mtb secretes over 70 serine proteases that may be used as viable drug targets. Hydrolase important for pathogenesis 1 (Hip1) is a serine protease found in Mtb with proteolytic activity that is required for dampening the host inflammatory response. Given the role of Hip1 in Mtb virulence, analyzing Hip1 substrate specificity is a critical step in designing selective inhibitors for effective tuberculosis treatment.

Fluorogenic substrates are nonfluorescent compounds that release a fluorescent reporter group when acted upon by an enzyme. Such substrates are useful biomolecular imaging tools that allow monitoring of enzymatic activity *in vitro* and *in vivo*. A large library of peptides was screened against Hip1, and enzymatic cleavage was analyzed using mass spectrometry to facilitate the design of fluorogenic substrates (Figure 1). The cleavage reaction has a large, negative ΔG° under the conditions used.

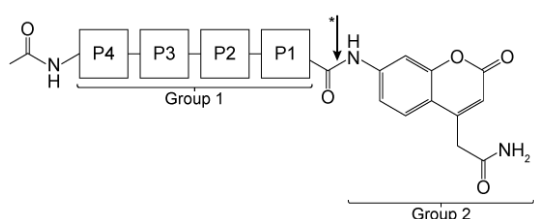


Figure 1 Schematic design of Hip1 fluorogenic substrates. P1, P2, P3, and P4 in Group 1 represent the positions of variable amino acids in the peptide. Group 2 depicts the fluorescent compound released on Hip1 cleavage.

Researchers reported Michaelis–Menten kinetic parameters for substrates from two peptide families: the WKLL substrates and the CSL substrates.

Table 1 Kinetic Parameters of Fluorogenic Substrates for Mtb Hip1

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Which of the following substrates yields the most optimal catalytic turnover and catalytic efficiency, respectively?

- ☐ A. CSL-173 and WKLL-ACC [8%]
- ☐ B. WKLL-ACC and CSL-176 [7%]
- ☐ C. CSL-176 and CSL-173 [12%]
- ☒ D. CSL-173 and CSL-176 [70%]

Correct

70% answered correctly

Explanation:

Substrate	k_{cat} (s ⁻¹)	K_m (M)	k_{cat}/K_m (s ⁻¹ M ⁻¹)
WKLL-ACC	1	8.3×10^{-5}	1.2×10^4
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Highest k_{cat} (turnover)

Highest k_{cat}/K_m ratio (efficiency)

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Catalytic turnover refers to the rate at which an enzyme-bound substrate is converted to product. It is given by the **rate constant** k_{cat} and is proportional to the **maximum reaction rate** V_{max} . On a Michaelis-Menten plot, catalytic turnover is reflected in the **height of the plateau** that occurs at high substrate concentrations. A high k_{cat} (or a high plateau) corresponds to high catalytic turnover.

Catalytic efficiency is defined by the ratio $\frac{k_{\text{cat}}}{K_m}$ and is a measure of enzyme specificity. It indicates the **rate of catalysis at low substrate** concentrations and is proportional to the **initial slope** of the Michaelis-Menten plot. An enzyme that quickly creates product at low substrate concentrations (high initial slope) is said to be catalytically efficient because only a small amount of substrate is needed to generate substantial activity. A high k_{cat} , a low K_m , or both may contribute to catalytic efficiency.

An enzyme-substrate system may have **any combination** of high and low efficiency and turnover. In this case, CSL-173 has the highest k_{cat} (turnover) and CSL-176 has the highest $\frac{k_{\text{cat}}}{K_m}$ ratio (efficiency).

(Choice A) Although CSL-173 has the highest catalytic turnover (largest k_{cat}), WKLL-ACC has the lowest catalytic efficiency ($\frac{k_{\text{cat}}}{K_m}$ ratio).

(Choice B) CSL-176 is the most efficient substrate, but WKLL-ACC has the lowest turnover.

(Choice C) This option has the optimal catalytic turnover and catalytic efficiency reversed.

Educational objective:

Catalytic turnover is represented by k_{cat} and is proportional to the maximum reaction velocity, which is obtained at high substrate concentrations. Catalytic efficiency is represented by $\frac{k_{\text{cat}}}{K_m}$

and reflects the ability of the enzyme to work at low substrate concentrations.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402113
System:
Enzymes

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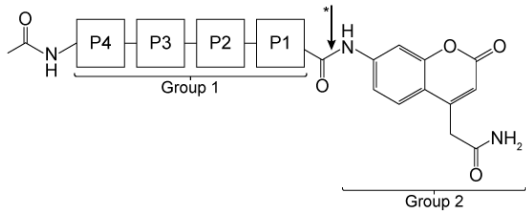


Figure 1 Schematic design of Hip1 fluorogenic substrates. P1, P2, P3, and P4 in Group 1 represent the positions of variable amino acids in the peptide. Group 2 depicts the fluorescent compound released on Hip1 cleavage.

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In another experiment, researchers assessed peptides of various compositions for enzymatic cleavage to survey the substrate preferences of Hip1. The results are shown in the table below.

PositionPreferred residue composition

- P4 Aliphatic
- P3 Basic
- P2 Aromatic
- P1 Polar uncharged

Going from P4 to P1, which sequence would be most suitable for the design of a selective Hip1 substrate?

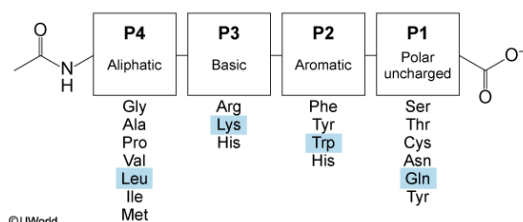
✔ ☒ A. Leu-Lys-Trp-Gln [70%]

- ☐ B. Ala-Arg-Leu-Asn [2%]
- ☐ C. Val-Gly-Phe-Ser [9%]
- ☐ D. Ile-Lys-Tyr-Trp [17%]

Correct

70% answered correctly

Explanation:



The variable side chain properties of amino acids enable diverse functions of peptides and proteins. Classification of amino acids based on their chemical properties is a useful way to understand protein interactions with other molecules. Amino acids can be broadly split into two categories, **hydrophilic** and **hydrophobic**. Hydrophilic amino acids tend to interact well with each other and with water whereas hydrophobic amino acids interact with each other and exclude water.

Each broad category can be further subdivided:

- Hydrophilic amino acids may be **polar but uncharged** (S, T, C, N, Q, Y), **acidic** (negatively charged at pH 7.4; D, E), or **basic** (positively charged at pH 7.4; K, R, H). Note that although tyrosine (Y) has a polar OH group, its large aromatic ring results in it often being classified as hydrophobic instead.
- Hydrophobic amino acids may be **aliphatic** (non-aromatic hydrocarbons; G, A, P, V, L, I, M) or **aromatic** (benzene ring forms; F, W). Note that **tyrosine** and **histidine**, which are polar, also contain aromatic rings.

The table indicates that the amino acids must be aliphatic, basic, aromatic, and polar uncharged. The peptide **Leu-Lys-Trp-Gln** satisfies these criteria, respectively.

(Choice B) This peptide contains leucine (Leu) at P2. Leucine is aliphatic, but the peptide requires an aromatic residue at P2.

(Choice C) This peptide contains glycine (Gly) at P3. Glycine is aliphatic, but the peptide requires a basic residue at P3.

(Choice D) This peptide contains tryptophan (Trp) at P1. Tryptophan is hydrophobic and aromatic, but the peptide requires a polar uncharged residue at P1.

Educational objective:

Grouping amino acids based on the properties of their side chains aids in understanding protein structure and function. Amino acids can generally be classified as hydrophobic or hydrophilic, and can be further subdivided into aliphatic, aromatic, polar uncharged, acidic, and basic.

The recent advent of multidrug-resistant *Mycobacterium tuberculosis* (Mtb) has led to the search for novel agents to combat tuberculosis. Mtb secretes over 70 serine proteases that may be used as viable drug targets. Hydrolase important for pathogenesis 1 (Hip1) is a serine protease found in Mtb with proteolytic activity that is required for dampening the host inflammatory response. Given the role of Hip1 in Mtb virulence, analyzing Hip1 substrate specificity is a critical step in designing selective inhibitors for effective tuberculosis treatment.

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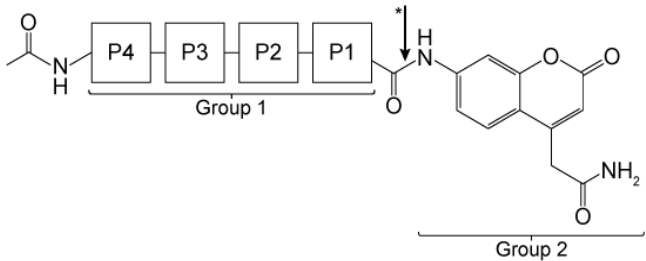


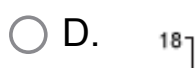
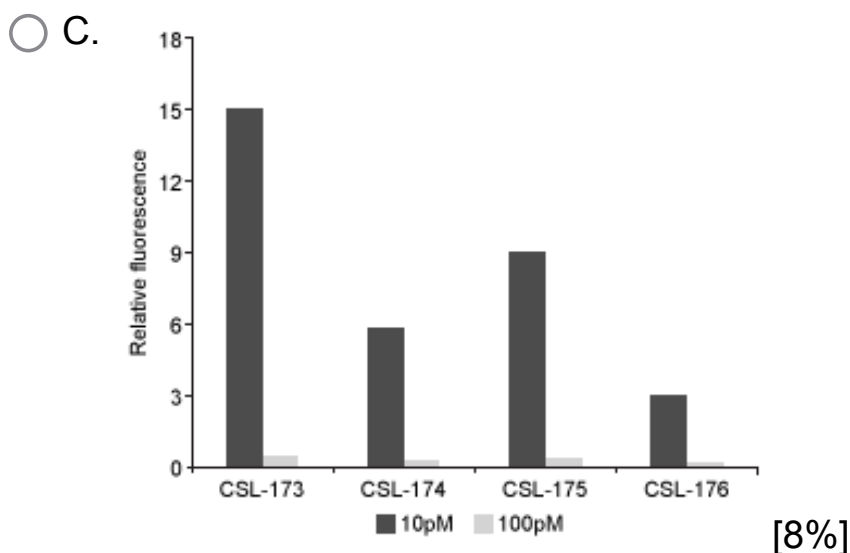
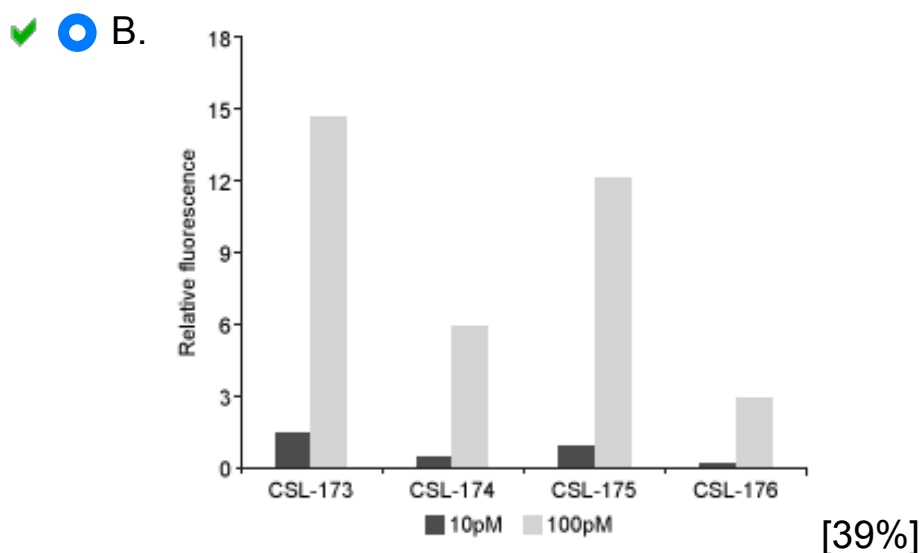
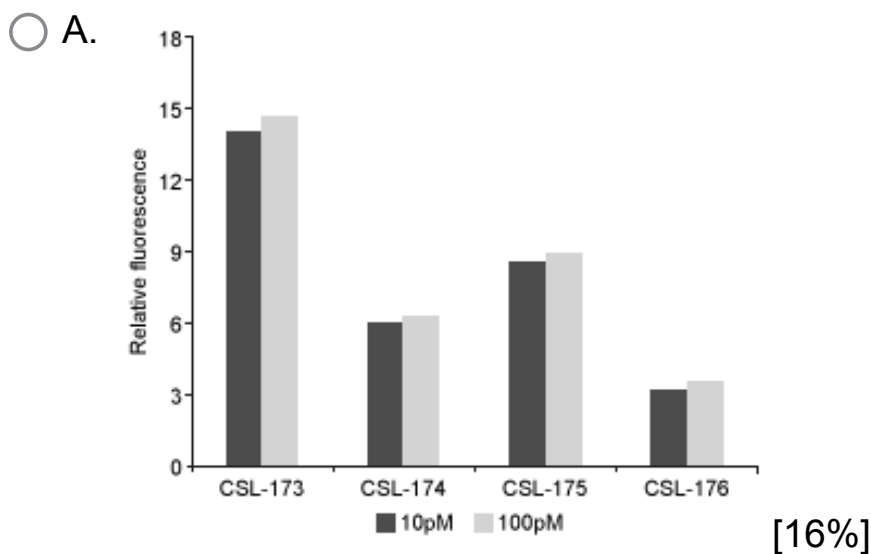
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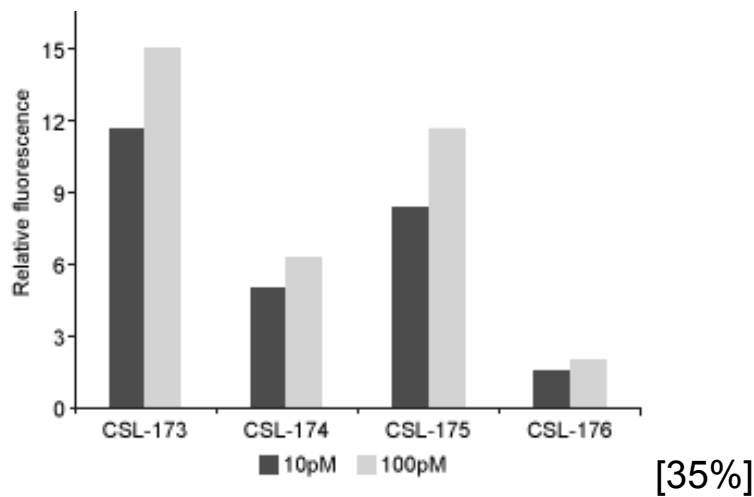
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The CSL substrates from Table 1 were tested *in vitro* at saturating substrate concentration (10 mM) in solutions containing 10 pM or 100 pM Hip1 enzyme. The reaction was allowed to run for 10 minutes and the relative fluorescence of each reaction vessel was then measured. Which of the following graphs best reflects the expected results?

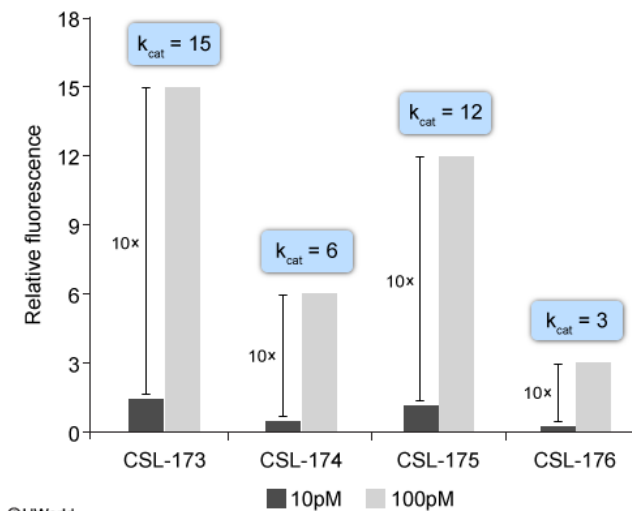




Correct

40% answered correctly

Explanation:



The rate of an enzyme-catalyzed reaction generally depends on the concentration of the enzyme $[E]$, the concentration of the substrate $[S]$, the catalytic turnover number k_{cat} , and the Michaelis constant K_m . The question states that the reaction is carried out with 10 mM substrate, which is many times greater than the largest K_m given in Table 1 (ie, saturating conditions). Under saturating conditions, enzymes essentially operate at their **maximum velocity V_{max}** .

V_{max} is directly proportional to k_{cat} , so the rate of catalysis under saturating conditions depends directly on k_{cat} . Therefore, the substrate with the highest k_{cat} (CSL-173) will be cleaved the fastest, will release the most fluorophores within the given 10-minute time span, and therefore will show the largest relative fluorescence (note that based on the information in the question and

Table 1, 10 minutes is **not enough time** to reach equilibrium). Similarly, the substrate with the second-highest k_{cat} (CSL-175) will produce the second-highest fluorescence in the given 10 minutes, and so forth.

V_{max} is also directly proportional to the **concentration of enzyme [E]**. Therefore, the fluorescence generated with 100 pM Hip1 enzyme should be **ten times greater** than the fluorescence generated with 10 pM Hip1 enzyme. Only one graph (**Choice B**) has both the correct relative intensity for each substrate *and* a proportional increase in fluorescence when tested with a greater concentration of enzyme.

(Choices A and D) These graphs do not show a proportional increase in enzymatic activity when enzyme concentration is increased.

(Choice C) This graph depicts a 10-fold increase in activity with 10 pM enzyme relative to 100 pM enzyme. This is the opposite of the expected trend.

Educational objective:

Under saturating conditions ($[S] \gg K_m$), enzymatic activity is directly proportional to k_{cat} because the enzyme is operating at its maximum velocity V_{max} . Because V_{max} is also directly proportional to enzyme concentration $[E]$, an increase in $[E]$ will proportionally increase V_{max} and therefore increase the total product output.

Time spent:0 sec

Subject:
Biochemistry

QID:402115

System:
Enzymes

The recent advent of multidrug-resistant *Mycobacterium tuberculosis* (Mtb) has led to the search for novel agents to combat tuberculosis. Mtb secretes over 70 serine proteases that may be used as viable drug targets. Hydrolase important for pathogenesis 1 (Hip1) is a serine protease found in Mtb with proteolytic activity that is required for dampening the host inflammatory response. Given the role of Hip1 in Mtb virulence, analyzing Hip1 substrate specificity is a critical step in designing selective inhibitors for effective tuberculosis treatment.

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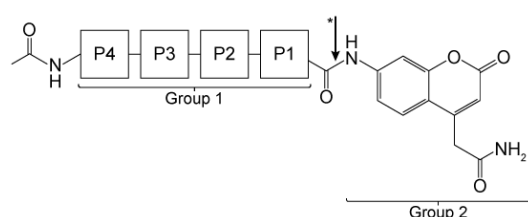


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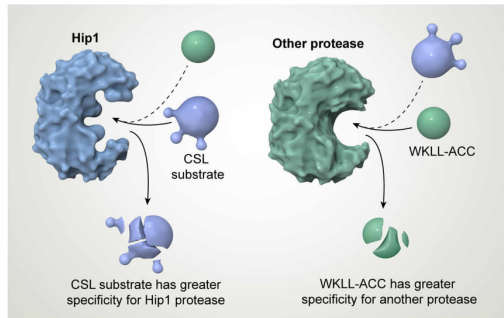
In further experiments researchers found that bacteria lacking Hip1 (knockout bacteria) were able to cleave WKLL-ACC but not the CSL substrates. How might these results be explained?

- ☐ A. WKLL-ACC readily undergoes hydrolysis in the absence of an enzyme. [27%]
- ☐ B. Knockout bacteria take up WKLL-ACC more readily than other substrates. [17%]
- ☒ C. WKLL-ACC has a greater specificity for another protease. [46%]
- ☐ D. Knockout bacteria have increased nonspecific protease activity. [8%]

Correct

46% answered correctly

Explanation:



The passage mentions that over 70 serine proteases are secreted by Mtb. WKLL-ACC has fairly low Hip1 specificity, as noted by the small $\frac{k_{\text{cat}}}{K_m}$ value (**catalytic efficiency**), but it may have **specificity** for one of the other proteases. Genetic removal of Hip1 would be a simple way to test which substrates require Hip1 and which work with other proteases.

Unlike WKLL-ACC, the CSL substrates have larger catalytic efficiency values and are also not strongly cleaved without Hip1. Because of this, the CSL substrates are most likely specific to Hip1. In contrast, the removal of Hip1 had a minimal effect on WKLL-ACC cleavage, suggesting that WKLL-ACC strongly interacts with one or more other proteases but does not interact strongly with Hip1.

(Choice A) Although peptide hydrolysis is a thermodynamically favorable process, it is kinetically slow. It does not occur quickly unless an enzyme is present to increase the rate.

(Choice B) The serine proteases discussed in the passage are secreted and function outside the cells, so peptide uptake into bacteria is irrelevant to the rate of hydrolysis.

(Choice D) Knocking one gene down can indeed result in upregulation of other genes to compensate. However, if the upregulated proteases were nonspecific, they would likely act on the CSL substrates just as readily as they would on WKLL-ACC.

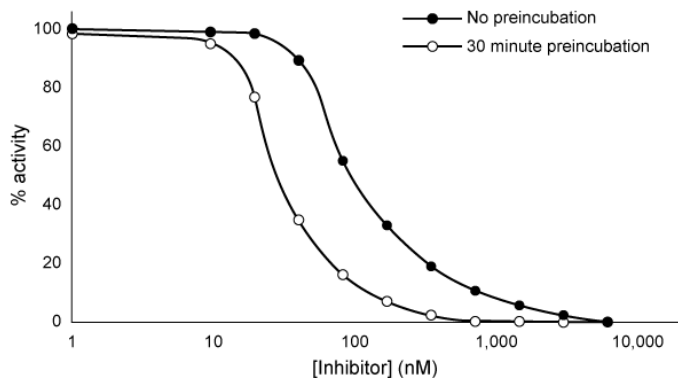
Educational objective:

Substrates with high specificity for one enzyme may have low specificity for another. Conversely, a substrate with low specificity for an enzyme of interest may be significantly acted upon by another enzyme in a living system. This scenario can be tested by knocking out a gene of interest in a living organism.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402116
System:
Enzymes

The graph below shows the effect of an inhibitor on enzyme activity when the enzyme and inhibitor were preincubated prior to the addition of a substrate and when no preincubation occurred. Constant amounts of enzyme and initial substrate concentration were used for each test. Based on the graph, which of the following statements is true?

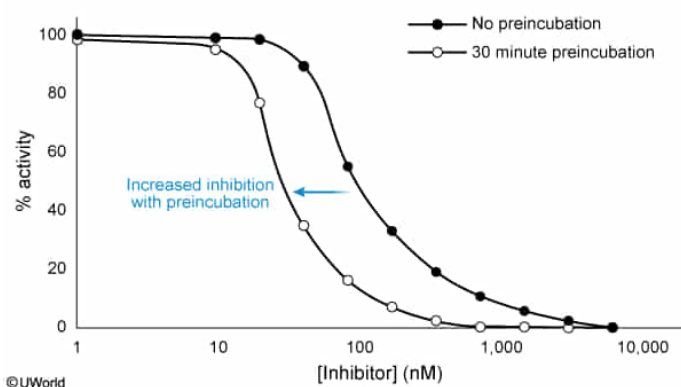


- ☐ A. The inhibitor most likely binds the enzyme reversibly. [28%]
- ☐ B. The inhibitor is more effective without preincubation. [10%]
- ☒ C. The inhibitor most likely binds the enzyme irreversibly. [51%]
- ☐ D. The inhibitor forms weak bonds with active site residues. [10%]

Correct

51% answered correctly

Explanation:



Irreversible inhibitors typically react with **nucleophilic side chains**, forming **covalent bonds to the enzyme**. Once covalently linked, the inhibitor **cannot be displaced** by any amount of substrate.

Preincubation with inhibitor provides **more time for covalent linkages to occur**; therefore, it can increase the level of inhibition because more

enzymes are inactivated. On the other hand, **reversible inhibitors** bind their target enzymes **noncovalently**. These binding interactions are generally much faster than the formation of covalent bonds, so additional time is not needed for reversible binding to occur. Therefore, reversible inhibitors tend to have the same effect whether they are preincubated with an enzyme or not.

With only 1 nM inhibitor, little change in activity was seen, indicating that such a low concentration of inhibitor is unable to effectively alter activity, even with preincubation. With ~20 nM inhibitor, little change was seen *without* preincubation (filled circles); however, preincubation (empty circles) allows for a **significant increase in inhibition**. This trend continues with higher concentrations of inhibitor, implying **inhibition is more effective with preincubation**. Because preincubation allows time for covalent linkages to occur, the data are most consistent with **an irreversible inhibitor**.

(Choice A) Inhibition increases when both inhibitor and enzyme are preincubated prior to activity assays. This indicates that binding is *irreversible*.

(Choice B) Less inhibitor is required to inactivate the enzyme when preincubation occurs, so the inhibitor is more effective *with* preincubation.

(Choice D) The inhibitor is most likely irreversible. To be irreversible, an inhibitor must form *strong* (usually covalent) bonds with active site residues.

Educational objective:

Irreversible inhibitors form covalent bonds with enzymes and become more potent given sufficient time to react (ie, with preincubation). Reversible inhibitors quickly form noncovalent bonds with target enzymes and do not require much time to achieve their full effect.

Time spent:0 sec
Subject:
Biochemistry

QID:402117
System:
Enzymes